

SANSH GOEL

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EDUCATION

University of Colorado, Boulder
Bachelor of Science in Computer Science

August 2022 – May 2026
GPA: 3.6/4.0

EXPERIENCE

University of Colorado, Boulder

August 2025 – December 2025

Undergraduate Course Assistant – Machine Learning & AI (CSCI 4622, CSCI 3202)

Boulder, CO

- Mentored **200+** students across **CSCI 4622: Machine Learning** and **CSCI 3202: Artificial Intelligence**, owning office hours, assignment support, and course logistics across two concurrent upper-division courses
- Debugged student **Python implementations** of **ML pipelines, AI algorithms, and data structures**, teaching systematic diagnosis of training errors, overfitting, and evaluation metric failures rather than just providing fixes
- Broke down complex concepts including **gradient descent, model evaluation, unsupervised learning, and search algorithms**, adapting explanations in real time to diverse backgrounds and bridging the gap between theory and working code
- Collaborated with **instructors and a cross-functional TA team** to align on grading standards, course improvements, and student support strategies across multiple semesters

HealthTech Solutions

May 2025 – August 2025

Software Engineer Intern — Data Engineering & Analytics

Denver, CO

- Architected an **AWS S3 → Snowflake** ingestion pipeline processing **5,000+** **healthcare records**, building a reliable data foundation that gave the client a **single source of truth** for business-critical reporting
- Optimized **20+ Snowflake SQL queries**, reducing latency enough to **enable real-time KPI tracking** — replacing batch reporting workflows that had previously delayed decision-making by hours
- Delivered **5+ Power BI dashboards** with automated refresh workflows used directly by **C-suite executives**, replacing manual reporting and providing live visibility into operational healthcare metrics

Salezshark

May 2024 – August 2024

Software Developer Intern — Frontend Engineering

New Delhi, India

- Refactored the company's **marketing platform in React.js**, implementing **code splitting and lazy loading** to reduce page load time by **30%**, directly improving user experience and search engine ranking performance
- Designed and implemented a **CI/CD pipeline in Azure DevOps**, automating build, test, and deployment workflows — cutting release cycles by **20%** and eliminating **15+ hours of manual deployment work** per sprint across the dev team
- Owned frontend feature development end-to-end across **8 Agile sprints**, driving **Git branching strategies and peer code reviews** within a lean 3-person dev team to ship clean, production-ready code under fast iteration pressure

PROJECTS

Savant — AI Research Assistant

MLH Hackathon — MongoDB Track Winner

Tech: Next.js, FastAPI, MongoDB Atlas Vector Search, Gemini API, Python, TypeScript, Chrome Extension, ElevenLabs, Vultr

- Won the **MLH MongoDB track** by building an AI research assistant that applied **Retrieval-Augmented Generation** to scientific PDFs using **Gemini embeddings** and **MongoDB Atlas Vector Search** to deliver citation-grounded answers
- Engineered an end-to-end document intelligence pipeline with **page-aware PDF parsing**, chunking, dense vector retrieval, **lexical fallback**, and custom reranking — containerized with **Docker** and deployed on **Vultr cloud** with latency telemetry across embedding, retrieval, and generation stages
- Developed **LLM-powered paper understanding** features including concept graph extraction, graph-based follow-up Q&A, and use-case synthesis, deployed through a **Next.js frontend**, **FastAPI backend**, and **Chrome extension** with voice I/O

Betty Boulder — Full-Stack Real Estate Platform

Capstone Project, In Development

Tech: Go, Gin, PostgreSQL, GORM, Redis, Docker, REST APIs, RentCast API

- Engineered **RESTful backend services in Go (Gin)** for a dual-portal real estate platform, powering property search, listing management, and **role-based access control** for agents and property seekers, containerized with **Docker** for consistent deployment
- Designed **relational data models with upsert logic** in PostgreSQL, implementing **conflict-safe writes and partial unique indexes** to maintain duplicate-free records across an automated **RentCast ingestion pipeline** synced via scheduled cron jobs

Machine Learning NLP Pipeline for Email Classification

Tech: Python, Scikit-learn, TF-IDF, K-Means, NLP, Pandas, NumPy

- Built an **end-to-end NLP pipeline** on a Kaggle email dataset — applying **TF-IDF vectorization → K-Means clustering → sentiment analysis** to transform raw unstructured text into labeled thematic categories
- Surfaced actionable **communication patterns and sentiment trends** across email clusters using **unsupervised ML**, achieving meaningful topic separation without labeled training data

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, Java, Go, C/C++

AI/ML: Model Evaluation, NLP Pipelines, Vector Similarity Search, Retrieval-Augmented Generation (RAG), Embedding Pipelines, Retrieval Ranking, Document Chunking, Scikit-learn, PyTorch, HuggingFace Transformers, LangChain

Data & Backend: FastAPI, PostgreSQL, MongoDB, Snowflake, AWS S3, ETL Pipelines, Data Warehousing, Data Modeling, Distributed Data Pipelines, REST APIs

Frontend & DevOps: React.js, Next.js, Docker, Azure DevOps, CI/CD, Git